DBD281 Project

Jason Beckley 602089, Gert-Pieter Van Zyl 604008, Lance Carolissen 605105,
Justin van Rooyen 604494

2026 FIFA World Cup Database Design

Table of Contents

Background of Organisation and Problem Statement……3

Business Rules…….4

Normalisation Process……..5

Entity Relationship Diagram……..14

Background of Organisation and Problem Statement

Background of Organisation

The organisation committee of the 2026 FIFA world cup has requested a database system to manage information about their tournament. They manage and coordinate large scale tournaments and oversee operational aspects such as team and ticket management. Without a sufficient database system, their decision making and data collected may be inaccurate and very difficult to manage.

Problem Statement

Without a database system to properly manage the data collected from FIFA tournaments, the data will be extremely messy, have huge redundancies and be very difficult to manage and use in decision making processes. The database system drawn out here aims to fix this, and will allow for better scalability, querying, reduced redundancy and error, and the ability to make sufficient use of the data collected from the FIFA tournaments.

Business Rules

Rule 1: Each team must represent one country

Rule 2: Each team must register between 23-26 players for the tournament

Rule 3: Each match must be played at exactly 1 stadium

Rule 4: Each player can only be registered to one team

Rule 5: A team cannot play against itself

Rule 6: Each stadium can only be located at 1 city

Rule 7: Each host city must be located in either Mexico, USA or Canada

Rule 8: Each match must involve 2 teams

Rule 9: Each match must have info Date, Time and Match Stage

Rule 10: Each ticket must be in reference to one match and one fan

Rule 11: Each fan can book multiple tickets

Rule 12: Each Stadium can host multiple matches

Rule 13: Each team must have at least one coach staff member

Rule 14: Each coaching staff member may only be assigned to one team

Rule 15: The total number of tickets bookings per match must not exceed the capacity of the stadium in which the match is played

Normalisation Process

UNF (Unnormalised Form)

TOURNAMENT DATA (

MatchID,

MatchDate,

MatchTime,

MatchStage

StadiumName,

StadiumCapacity,

City,

HostCountry,

HomeTeamName,

HomeTeamCountry,

HomeTeamPlayer1_Name, HomeTeamPlayer1_Jersey, HomeTeamPlayer1_Position, HomeTeamPlayer1_DOB,

HomeTeamPlayer2_Name, HomeTeamPlayer2_Jersey, HomeTeamPlayer2_Position, HomeTeamPlayer2_DOB,

… (continues)

HomeTeamCoach1_Name, HomeTeamCoach1_Role,

HomeTeamCoach2_Name, HomeTeamCoach2_Role,

…(continues)

HomeScore,

AwayTeamName,

AwayTeamCountry,

AwayTeamPlayer1_Name, AwayTeamPlayer1_Jersey, AwayTeamPlayer1_Position, AwayTeamPlayer1_DOB,

AwayTeamPlayer2_Name, AwayTeamPlayer2_Jersey, AwayTeamPlayer2_Position, AwayTeamPlayer2_DOB,

…(continues)

AwayTeamCoach1_Name, AwayTeamCoach1_Role,

AwayTeamCoach2_Name, AwayTeamCoach2_Role,

…(continues)

AwayScore,

Ticket1_ID, Ticket1_FanName, Ticket1_FanEmail, Ticket1_SeatNumber, Ticket1_Section, Ticket1_Price,

Ticket2_ID, Ticket2_FanName, Ticket2_FanEmail, Ticket2_SeatNumber, Ticket2_Section, Ticket2_Price,

…(continues)

Attendance

)

1NF (First Normal Form) – Repeating groups removed

TOURNAMENT DATA (

MatchID,

MatchDate,

MatchTime,

MatchStage,

StadiumName,

StadiumCapacity,

City,

HostCountry,

HomeTeamName,

HomeTeamCountry,

AwayTeamName,

AwayTeamCountry,

HomeScore,

AwayScore,

TeamName,

PlayerName,

PlayerJerseyNumber,

PlayerPosition,

PlayerDateOfBirth,

CoachName,

CoachRole,

FanName,

FanEmail,

FanPhone

TicketID,

SeatNumber,

Section,

TicketPrice,

PurchaseDate

Attendance

)

Composite PK (MatchID, TeamName, PlayerName, CoachName, TicketID)

2NF (Second Normal Form) – Elimination of partial dependencies

MATCH (

MatchID, (PK)

MatchDate,

MatchTime,

MatchStage,

StadiumName,

HomeTeamName,

AwayTeamName,

HomeScore,

AwayScore

Attendance

)

STADIUM (

StadiumName, (PK)

Capacity,

City

)

TEAM (

TeamName, (PK)

Country

)

PLAYER (

PlayerName, (PK)

TeamName, (FK – TEAM)

JerseyNumber,

Position,

DateOfBirth

)

COACHING_STAFF (

CoachName, (PK)

TeamName, (FK – TEAM)

Role

)

FAN (

FanName, (PK)

Email,

Phone

)

TICKET (

TicketID, (PK)

MatchID, (FK - MATCH)

FanName, (FK - FAN)

SeatNumber,

Section,

Price,

PurchaseDate

3NF (Third Normal Form) – Elimination of transitive dependencies (Also using Ids, as well as declaring types)

COUNTRY (

CountryID, --INT (PK)

CountryName --NVARCHAR(100) NOT NULL, UNIQUE

)

CITY (

CityID, --INT (PK)

CityName, --NVARCHAR(100) NOT NULL

CountryID --INT NOT NULL (FK - COUNTRY)

)

STADIUM (

StadiumID, --INT (PK)

StadiumName, --NVARCHAR(150) NOT NULL

Capacity, -- INT NOT NULL

CityID --INT NOT NULL (FK – CITY)

)

TEAM (

TeamID, --INT (PK)

TeamName, --NVARCHAR(100) NOT NULL, UNIQUE

CountryID --INT NOT NULL (FK-COUNTRY)

)

```
PLAYER (
PlayerID,          --INT (PK)
FirstName,      -- NVARCHAR(50) NOT NULL
LastName,      --NVARCHAR(50) NOT NULL
DateOfBirth,    --DATE NOT NULL
Position,          --NVARCHAR(20) NOT NULL
JerseyNumber,   --INT NOT NULL
TeamID      --INT NOT NULL (FK – TEAM)
UNIQUE (TeamID, JerseyNumber)
)

COACHING_STAFF (
CoachID, --INT (PK)
FirstName,      --NVARCHAR(50) NOT NULL
LastName,    --NVARCHAR(50) NOT NULL
Role,       --NVARCHAR(50) NOT NULL
TeamID     --INT NOT NULL   (FK-TEAM)
)
MATCH (
MatchID,          --INT (PK)
MatchDate,      --DATE NOT NULL
MatchTime,     --TIME NOT NULL
MatchStage,     --NVARCHAR(50) NOT NULL
StadiumID,       --INT NOT NULL  (FK-STADIUM)
HomeTeamID,     --INT NOT NULL  (FK-TEAM)
AwayTeamID,  --INT NOT NULL  (FK-TEAM)
HomeScore,     --INT DEFAULT 0
AwayScore      --INT DEFAULT 0
```

Attendance --INT DEFAULT 0

)

FAN (

FanID, --INT (PK)

FirstName, --NVARCHAR(50) NOT NULL

LastName, --NVARCHAR(50) NOT NULL

Email, --NVARCHAR(100) NOT NULL, UNIQUE

Phone, --NVARCHAR(20)

CountryID --INT NOT NULL (FK-COUNTRY)

)

TICKET (

TicketID, --INT (PK)

MatchID, --INT NOT NULL (FK-MATCH)

FanID, --INT NOT NULL (FK-FAN)

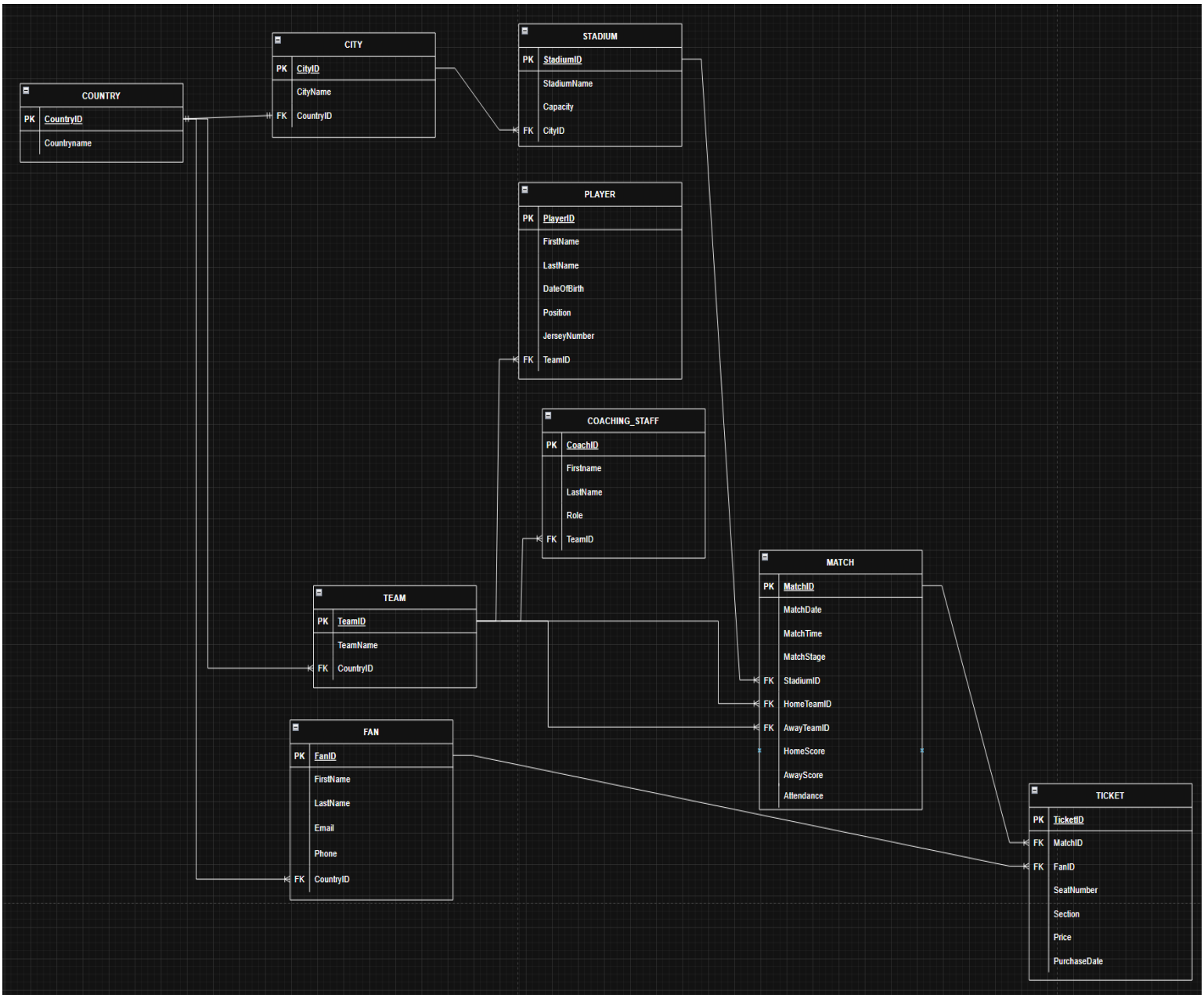
SeatNumber, --NVARCHAR(10) NOT NULL

Section, --NVARCHAR(50)

Price, --DECIMAL(10,2) NOT NULL

PurchaseDate DATETIME NOT NULL

UNIQUE (MatchID, SeatNumber)

)

Entity Relationship Diagram